

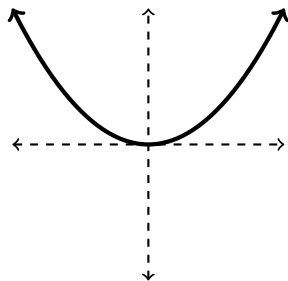
MATH 195: 9/15

POWER FUNCTIONS

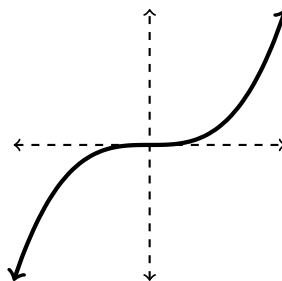
A *power function* is one of the form $f(x) = x^n$ for n a positive integer. We also will call their geometric transformations power functions, rather than awkwardly say “geometric transformations of power functions”.

The general shape of the graph of x^n depends on the *parity* of n .

n is even.



n is odd ($n \geq 3$).



Vertex at $(0,0)$. This is where the graph changes between decreasing and increasing. It is concave up everywhere.

Inflection point at $(0,0)$. This is where the *concavity* of the graph changes. It is increasing everywhere.

- (1) Explain why x^n for even n all look like the graph on the left.
- (2) Sketch graphs of x^2 and x^4 on the same axes. What is different about them?
- (3) Explain why x^n for odd n all look like the graph on the right.
- (4) Sketch graphs of x^3 and x^5 on the same axes. What is different about them?
- (5) Why exclude $n = 1$ from the picture on the right? In other words, why does x^1 look different?

When sketching a graph of a geometric transformation of a power function, there are two more important details:
 (1) Where is the vertex/inflection point? (2) What is the orientation? The first is answered by identifying the horizontal and vertical translations. The second is answered by determining whether there was a vertical reflection.

- (1) Explain why $a(x) = (x - 2)^3 - 1$ has its inflection point at $(2, -1)$. Sketch a graph.
- (2) What is the orientation of $b(x) = -(x + 1)^4$? Sketch a graph, identifying its vertex.
- (3) Sketch a graph of $c(x) = -3(x - 3)^5 - 3$, identifying its inflection point. What is its domain? Range? Where is the graph increasing? Decreasing? Concave up? Concave down?
- (4) Sketch a graph of $d(x) = 2(x + 2)^4 - 3$, identifying its inflection point. What is its domain? Range? Where is the graph increasing? Decreasing? Concave up? Concave down?
- (5) Explain why a horizontal reflection on a power function of even parity doesn't change the graph. Give both a graphical and algebraic explanation.
- (6) Explain why a horizontal reflection on a power function of odd parity is equivalent to doing a vertical reflection. Give both a graphical and algebraic explanation.
- (7) Explain why a horizontal scaling on a power function is equivalent to a vertical scaling by a different amount. Determine the factor of the vertical scaling equivalent to a horizontal scaling by a . [Hint: your answer will depend upon n .]
- (8) Write, as simply as possible, a general form for a function which is a geometric transformation of a power function. Why does your form cover all six geometric transformations?
- (9) Explain why every power function of odd degree has $(-\infty, \infty)$ as its range.
- (10) Explain how to determine the range of a power function of even degree.
- (11) Explain how to determine where a power function is increasing, decreasing, concave up, and concave down. [Hint: you should have different answers depending on the parity of n .]
- (12) Explain how you find the intercepts of a power function. Create and solve an example.